Varunprasaath R. Selvaraju, Ph.D.

(+1) 678-207-9914 | varunprasaath@gmail.com | linkedin.com/in/varunprasaath-rs |

PROFESSIONAL SUMMARY

PhD organic/materials chemist with 6+ years designing, synthesizing, purifying, and characterizing small-molecule and polymeric materials through NMR, MS, GPC/SEC, FT-IR, UV-Vis-NIR, electrochemistry, TGA/DSC, and DOE/SPC. Connects synthetic strategy, analytical readouts, and structure-property data to data-rich discovery decisions and reproducible technical documentation.

CORE COMPETENCIES

Synthetic Organic Chemistry | Medicinal Chemistry Problem Solving | Purification & Characterization | NMR / LC-MS / GC-MS | Structure-Property Analysis | Spectroscopy & Electrochemistry | DOE/SPC Experimental Design | Scientific Publications

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided chemical process development for bio-based polymer systems, defining reaction/process variables, analytical readouts, reproducibility criteria, and scale-up specifications.
- Translate synthetic and process outcomes into SOP-style documentation, technical reports, and data packages that support iterative experimental planning and development decisions.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Designed, synthesized, purified, and characterized 15+ xanthene-based electrochromic molecular variants, linking substituent changes to redox behavior, optical response, stability, and material performance.
- Applied NMR, mass-spectrometry-informed analysis, GPC/SEC, FT-IR, UV-Vis-NIR, cyclic voltammetry, TGA, and DSC to confirm identity, assess purity, troubleshoot degradation, and prioritize next analogs.
- Built structure-property-stability datasets and publication-quality technical narratives for functional organic molecules, including work published in Chemistry of Materials.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Drove process development for CVD-based molecular thin-film coatings, optimizing surface preparation and deposition conditions to improve reproducibility, morphology, and optical performance.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Optimized CVD polymer thin-film deposition under flow using DOE and in-situ QCM-D monitoring to quantify growth kinetics and process controls.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, and safety systems that improved instrument uptime by 30%.

Georgia Tech 2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Electrochromic Molecular Design Chemistry of Materials 2025

Designed and evaluated 15+ organic molecular variants through synthesis, purification, analytical characterization, and property-driven iteration.

Organic synthesis, NMR, MS, FT-IR, UV-Vis-NIR, CV, GPC

Bio-Based Polymer Scale-Up Process Current

Built DOE-guided process windows, analytical readouts, and material-quality specifications for reproducible polymer/formulation scale-up.

DOE, SPC, process variables, analytical characterization

AFOSR Defense Thin-Film Coatings DoD Funded

Converted early-stage coating concepts into controlled CVD process conditions and reproducible thin-film outputs.

CVD, QCM-D, surface prep, process optimization

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Synthetic and Medicinal Chemistry: Organic synthesis, molecular design, purification/workup, structure-property relationships, physical organic reasoning, SAR-adjacent analog analysis
Analytical Characterization: NMR, GC-MS, LC-MS, GPC/SEC, FT-IR, UV-Vis-NIR, ellipsometry, cyclic voltammetry, spectroelectrochemistry, TGA, DSC, QCM-D
Data and Experimental Design: DOE, Statistical Process Control (SPC, Coursera 2026), JMP (basic), Python (basic), Excel (advanced), reproducibility analysis
Materials and Process Expertise: Organic electrochromic molecules, polymer systems, thin-film coatings, CVD deposition, surface functionalization, process optimization, degradation analysis
Scientific Communication: Technical reports, peer-reviewed publications, conference presentations, SOP-ready documentation, cross-functional technical updates

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)